

Loan Limit Optimization Analysis

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1. Problem Statement

The problem is to determine how to do loan limit optimization to maximize customer retention, lifetime value, and profitability.

1.1 Key Considerations

- **Dynamic credit eligibility** – borrowers move through different risk categories based on past behavior
- **Markov chain modeling** – predicting customer transitions between risk states
- **Stochastic demand forecasting** – estimating customer uptake of limit increases
- **Loan lifecycle simulation** – modeling multi-period borrowing behavior and default probabilities
- **Constraint optimization** – ensuring regulatory compliance and capital efficiency constraints are met

1.2 Constraint Optimization Framework

The optimization model incorporates multiple constraints to ensure practical implementability:

Portfolio Default Rate Constraint: The weighted average default rate across all risk tiers must not exceed a specified threshold (40%). This ensures the portfolio maintains acceptable risk levels while pursuing profitability.

Allocation Bounds: Each risk tier must receive a minimum allocation (5-10%) to maintain portfolio diversity, while maximum allocation caps (40-55%) prevent over-concentration in any single tier.

Capacity Constraint: Monthly processing capacity limits the total number of limit increases that can be executed, set at 16,000 based on operational throughput estimates.

Eligibility Requirements: Customers must meet minimum criteria (60 days since last increase) before being eligible for another limit increase, with a maximum of 6 increases per customer annually.

1.3 Given Parameters

Parameter	Value	Description
Profit per Increase	\$40	Net profit from each successful limit increase
Eligibility Threshold	60 days	Minimum days since last increase before customer is eligible
Max Increases per Year	6	Maximum limit increases allowed per customer annually
Annual Discount Rate	19%	Rate used for NPV calculations

1.4 Modeling Assumptions (Our Estimates)

The following parameters were assumed for the optimization model:

Parameter	Value	Rationale
Monthly Capacity	16,000	Based on demand forecast showing 13,300-17,000 monthly eligible customers
Max Portfolio Default Rate	40%	Risk appetite ceiling for optimization constraint
Loss Given Default	\$120	Assumed as 3x profit per increase
Min Tier Allocation	5-10%	Ensures all segments receive some allocation
Max Tier Allocation	40-55%	Prevents over-concentration in any single tier
Uptake Rates	60-85%	Estimated acceptance rates by risk tier (Low: 85%, High: 60%)

2. Data Overview

2.1 Dataset Characteristics

Metric	Value
Total Records	30,000
Number of Columns	6
Missing Values	0 (none)
Time Period	12 months of historical data

2.2 Data Columns

Column	Description
customer_id	Unique identifier for each customer
payment_segment	Customer payment behavior category
days_since_last_increase	Days elapsed since last limit increase
num_increases_ytd	Number of limit increases year-to-date
profit	Recorded profit from customer
default_flag	Binary indicator of customer default (0/1)

2.3 Data Quality Findings

Anomalies Detected:

- Profit Calculation Mismatch: 56% of records show a discrepancy between the recorded profit values and the expected calculation based on number of increases and profit per increase (\$40). This suggests additional factors or adjustments in the source data.

Critical Finding – Uniform Default Rate:

Analysis revealed that the default rate is approximately 40% uniformly across ALL payment segments. This indicates that the current lending policy does not effectively differentiate risk based on payment history. The existing segmentation provides no predictive power for default probability.

3. Methodology

3.1 Initial Approach: Exploratory Data Analysis

Objective: Identify patterns in historical data to predict default risk and optimize limit allocation.

Hypothesis: Payment behavior segments would show differentiated default rates, enabling risk-based limit allocation.

Findings: The EDA revealed that historical default rates are uniformly distributed (~50%) across all segments. Statistical tests confirmed no significant difference in default probability between payment segments. This means the current policy treats all customers similarly regardless of payment behavior.

Why Descriptive Analysis Failed: The uniform default rate across segments indicates that historical data cannot be used to build a predictive model—there is no signal to differentiate risk. The existing limit allocation policy is not risk-based, making historical patterns unsuitable for optimization.

3.2 Shift to Prescriptive Analytics

Given the limitations of descriptive analysis, we shifted to a prescriptive approach using theoretical risk models and operations research techniques. Rather than relying on historical patterns, we constructed a forward-looking optimization framework that:

- Defines risk tiers based on theoretical payment behavior characteristics
- Models customer transitions between risk states using Markov chains
- Forecasts demand under uncertainty using stochastic simulation
- Simulates loan lifecycles via Monte Carlo methods
- Optimizes allocation under realistic operational constraints

4. Module-by-Module

4.1 Module 1: Data Preprocessing and Exploratory Analysis

Objective: Clean data, identify patterns, and assess feasibility of predictive modeling.

Hypothesis: Historical payment patterns would correlate with default risk.

Calculations:

- Computed default rates by payment segment using contingency analysis
- Applied K-means clustering to identify natural customer groupings
- Calculated profit calculation match rates

Observations: No statistically significant difference in default rates across segments. Clustering analysis confirmed uniform distribution. Historical data lacks predictive power for risk differentiation, necessitating a theoretical model approach.

4.2 Module 2: Markov Chain Model for Risk State Transitions

Objective: Model customer risk dynamics over time using probabilistic state transitions.

Hypothesis: Customer risk profiles evolve predictably based on payment behavior, allowing forward-looking risk assessment.

Calculations:

- Constructed 4-state Markov chain (Low Risk → Medium-Low → Medium-High → High Risk)
- Estimated transition probability matrix from theoretical payment behavior
- Computed stationary distribution: Low 20.2%, Med-Low 28.3%, Med-High 28.7%, High 22.8%
- Derived monthly default probabilities by state: 2.4% to 5.6%
- Calculated expected time in each state: 4.1 to 9.8 months

Observations: The theoretical model shows meaningful risk differentiation. Low-risk customers have 2.4% monthly default probability vs. 5.6% for high-risk. Lifetime value ranges from \$136 (high risk) to \$224 (low risk), providing a basis for differentiated allocation strategies.

4.3 Module 3: Stochastic Demand Forecasting

Objective: Forecast customer demand for limit increases under uncertainty.

Hypothesis: Demand varies with economic conditions, seasonality, and customer risk profiles.

Calculations:

- Simulated 1,000 demand scenarios using geometric Brownian motion
- Modeled economic factors: inflation, unemployment, interest rates
- Applied seasonal adjustment factors (0.9 to 1.15 by month)
- Estimated uptake rates by risk tier: 45% to 70%
- Conducted stress testing under adverse scenarios

Observations: Monthly eligible customers range from 13,200 to 17,250. Peak demand occurs in November-December (seasonal factor 1.10-1.15). Demand is sensitive to economic conditions—a 2% inflation increase raises uptake by approximately 4%.

4.4 Module 4: Loan Lifecycle Simulation (Monte Carlo)

Objective: Simulate portfolio performance under various scenarios to quantify risk and return.

Hypothesis: Portfolio NPV and default rates can be reliably estimated through repeated simulation.

Calculations:

- Ran 100 Monte Carlo simulations over 12-month horizon
- Simulated individual customer journeys through risk states
- Calculated period-by-period profits and losses
- Applied 19% annual discount rate for NPV calculation
- Computed portfolio statistics: mean, std dev, percentiles

Observations:

Metric	Value
Mean Portfolio NPV	\$5,782,383
Standard Deviation	\$19,606
5th Percentile (VaR)	\$5,748,608
95th Percentile	\$5,816,429
Mean Default Rate	38.5%

The simulation shows stable portfolio performance with low variance (coefficient of variation ~0.3%). The 5th percentile NPV of \$5.75M provides a conservative estimate for planning purposes.

4.5 Module 5: Optimization Engine

Objective: Determine optimal allocation across risk tiers to maximize NPV while satisfying constraints.

Hypothesis: An optimal allocation exists that outperforms equal distribution across risk tiers.

Mathematical Formulation:

Maximize:

$$NPV = \sum_i (\text{allocation}_i \times \text{customers}_i \times \text{profit}_i \times (1 - \text{default}_i)) / (1 + r)^t$$

Subject to:

- $\sum_i \text{allocation}_i = 1$ (total allocation equals 100%)
- $\sum_i (\text{allocation}_i \times \text{default}_i) \leq \text{max_default_rate}$
- $\text{min_allocation} \leq \text{allocation}_i \leq \text{max_allocation}$ for all i
- $\sum_i (\text{allocation}_i \times \text{customers}_i) \leq \text{monthly_capacity}$

Observations:

Parameter	Optimal Value
Low Risk Allocation	40%
Medium-Low Risk Allocation	40%
Medium-High Risk Allocation	10%
High Risk Allocation	10%
Optimized NPV	\$5,558,354
Baseline NPV	\$5,385,990
NPV Improvement	\$172,364 (+3.2%)
Weighted Default Rate	40.0%

The optimizer converged in 2 iterations, concentrating allocation on lower-risk tiers (80% combined) while maintaining minimum exposure to higher-risk segments for portfolio diversity.

4.6 Module 6: Results and Reporting

Objective: Consolidate findings and generate actionable recommendations.

Outputs Generated:

- Executive dashboard visualization
- Risk assessment matrix by tier
- Implementation roadmap with 12-month timeline
- Sensitivity analysis of key parameters

5. Research Questions and Answers

5.1 What is the optimal loan limit increase strategy that balances profitability and risk?

The optimal strategy allocates 40% of capacity to Low Risk customers, 40% to Medium-Low Risk, and 10% each to Medium-High and High Risk tiers. This allocation achieves:

- Expected annual NPV of \$5.56 million
- Portfolio default rate of 40% (within 50% constraint)
- 3.2% improvement over equal allocation baseline

The strategy prioritizes lower-risk customers who generate higher expected lifetime value (\$224 vs \$136) while maintaining minimum allocation to all tiers for diversification.

5.2 What advanced operations research techniques best model this problem?

The analysis employs four complementary OR techniques:

- Markov Chains: Model customer risk state transitions with a 4-state transition matrix
- Stochastic Simulation: Geometric Brownian motion for demand forecasting with economic factors
- Monte Carlo Simulation: 100 portfolio scenarios for NPV distribution and risk metrics
- Constrained Optimization: SLSQP algorithm for allocation under multiple business constraints

5.3 How does borrower behavior impact the optimal strategy?

Borrower behavior affects the strategy through three channels:

Repayment patterns: Drive transitions between risk states. Consistent on-time payments move customers to lower-risk tiers over time.

Churn: Implicit in the Markov model through absorbing default states. Higher-risk customers have shorter expected tenure.

Limit acceptance: Uptake rates vary by tier (45-70%). Higher-risk customers show higher acceptance rates, but this is offset by higher default probability.

5.4 How do external economic conditions affect loan limit optimization?

Economic conditions impact the model through the demand forecasting module:

- Inflation (+2%): Increases uptake rates by ~4% as customers seek additional credit

- Unemployment (+1%): Decreases uptake by ~1.5% as eligible pool shrinks
- Interest rates (+1%): Decreases uptake by ~0.5% due to higher borrowing costs

Stress testing under adverse scenarios (high inflation + high unemployment) shows portfolio NPV remains positive but with increased variance. The optimization model is robust to moderate economic shocks.

5.5 What innovative strategies can enhance profitability while reducing default rates?

Several strategies emerge from the analysis:

Dynamic reallocation: Adjust tier allocations monthly based on observed transition rates and economic indicators.

Behavioral nudges: Encourage on-time payments through small incentives, accelerating transition to lower-risk tiers.

Tiered limit increases: Offer smaller increases to high-risk customers (\$200) vs larger increases to low-risk (\$500).

Seasonal timing: Concentrate offers during high-uptake periods (Nov-Dec) when seasonal factors exceed 1.10.

Reinforcement learning: Future enhancement using real-time customer response data to dynamically optimize individual offers.

6. Recommendations for Operationalization

6.1 Immediate Actions (Months 1-3)

- Implement risk-based segmentation using the 4-tier framework
- Deploy allocation engine with 40/40/10/10 split as initial configuration
- Establish monthly capacity monitoring at 16,000 increases
- Create dashboard tracking NPV, default rates, and tier distributions

6.2 Short-term Enhancements (Months 4-6)

- Integrate real-time payment behavior data for dynamic risk scoring
- Implement A/B testing framework to validate model predictions
- Develop early warning indicators for tier migration
- Refine uptake rate estimates based on actual acceptance data

6.3 Medium-term Optimization (Months 7-12)

- Calibrate Markov transition probabilities from observed customer behavior
- Implement automated rebalancing based on portfolio drift
- Develop customer-level propensity models for targeted offers
- Integrate economic indicators for demand adjustment

6.4 Key Performance Indicators

KPI	Target/Action
Monthly NPV	Track against \$463K monthly target
Portfolio Default Rate	Maintain below 50% threshold
Tier Allocation Compliance	Verify actual vs. target allocation
Capacity Utilization	Monitor against 16,000 monthly limit
NPV Variance	Track simulation accuracy vs. actuals

7. Conclusion

This analysis demonstrates that a structured, constraint-based optimization approach can improve loan limit allocation decisions by approximately 3.2% NPV over baseline strategies. The key insight is that historical data alone—showing uniform default rates across segments—cannot drive effective risk differentiation. Instead, a prescriptive framework combining Markov chain dynamics, stochastic demand modeling, and constrained optimization provides a rigorous foundation for allocation decisions.

The recommended 40/40/10/10 allocation across Low, Medium-Low, Medium-High, and High risk tiers balances profitability objectives with prudent risk management. This allocation is robust to moderate economic shocks and can be dynamically adjusted as real customer behavior data accumulates.

Implementation should proceed in phases, starting with the core allocation engine and progressively incorporating real-time data feeds and automated rebalancing. The framework is designed to evolve from theoretical parameters to empirically calibrated models as operational experience accumulates.